University Institute of Computing

Project Report

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Coffee Scale Analysis Report

1. Introduction

Coffee is one of the most widely consumed beverages globally, with a complex supply chain and diverse consumer preferences. Coffee scale analysis refers to evaluating coffee production, distribution, consumption, and quality across different scales—ranging from small farms to global markets.

2. Objectives

* To analyze coffee production at local, national, and global levels
* To evaluate quality parameters using measurement techniques
* To understand economic and environmental impacts
* To study demand trends and consumer behavior

3. Types of Coffee Production Scales

3.1 Small-Scale Production:

* Typically family-owned farms
* Found in countries like Ethiopia and Colombia
* Focus on specialty or organic coffee
* Limited mechanization

3.2 Medium-Scale Production:

* Cooperative farms or regional producers

* Moderate use of machinery
* Balanced focus on quality and quantity

3.3 Large-Scale Production:

* Industrial plantations
* Common in Brazil and Vietnam
* High output with mechanized processes
* Focus on export and mass production

4. Coffee Measurement and Scale Tools

4.1 Coffee Scales:

Precision scales are used to measure:

* Coffee beans (grams)
* Water-to-coffee ratio
* Brewing consistency

4.2 Brew Ratios:

* Standard ratio: 0 to 54(coffee:water)
* Impacts taste strength and extraction

4.3 Extraction Yield:

* Measures how much flavor is extracted
* Ideal range: 18%–22%

4.4 Total Dissolved Solids (TDS):

* Indicates coffee strength
* Measured using refractometers

5. Global Coffee Production Analysis

5.1 Production Trends

* Increasing demand worldwide
* Climate change affecting yields
* Shift toward specialty coffee markets

6. Economic Impact

* Coffee is a major export commodity
* Provides livelihood to millions of farmers
* Price fluctuations affect developing economies
* Influenced by global trade and market demand

7. Environmental Considerations

* Deforestation due to plantation expansion
* Water usage in processing
* Climate change affecting coffee-growing regions
* Sustainable farming practices gaining importance

8. Consumer Trends

* Rise of specialty coffee culture
* Increased demand for organic and fair-trade coffee
* Growth of café chains like Starbucks
* Preference for home brewing with precision tools

9. Challenges in Coffee Scale Analysis

* Inconsistent quality measurement
* Climate variability
* Market price instability
* Limited access to technology for small farmers

10. Conclusion

Coffee scale analysis provides valuable insights into production efficiency, quality control, and market trends. From small-scale farmers to global corporations, understanding these scales helps improve sustainability, profitability, and consumer satisfaction.

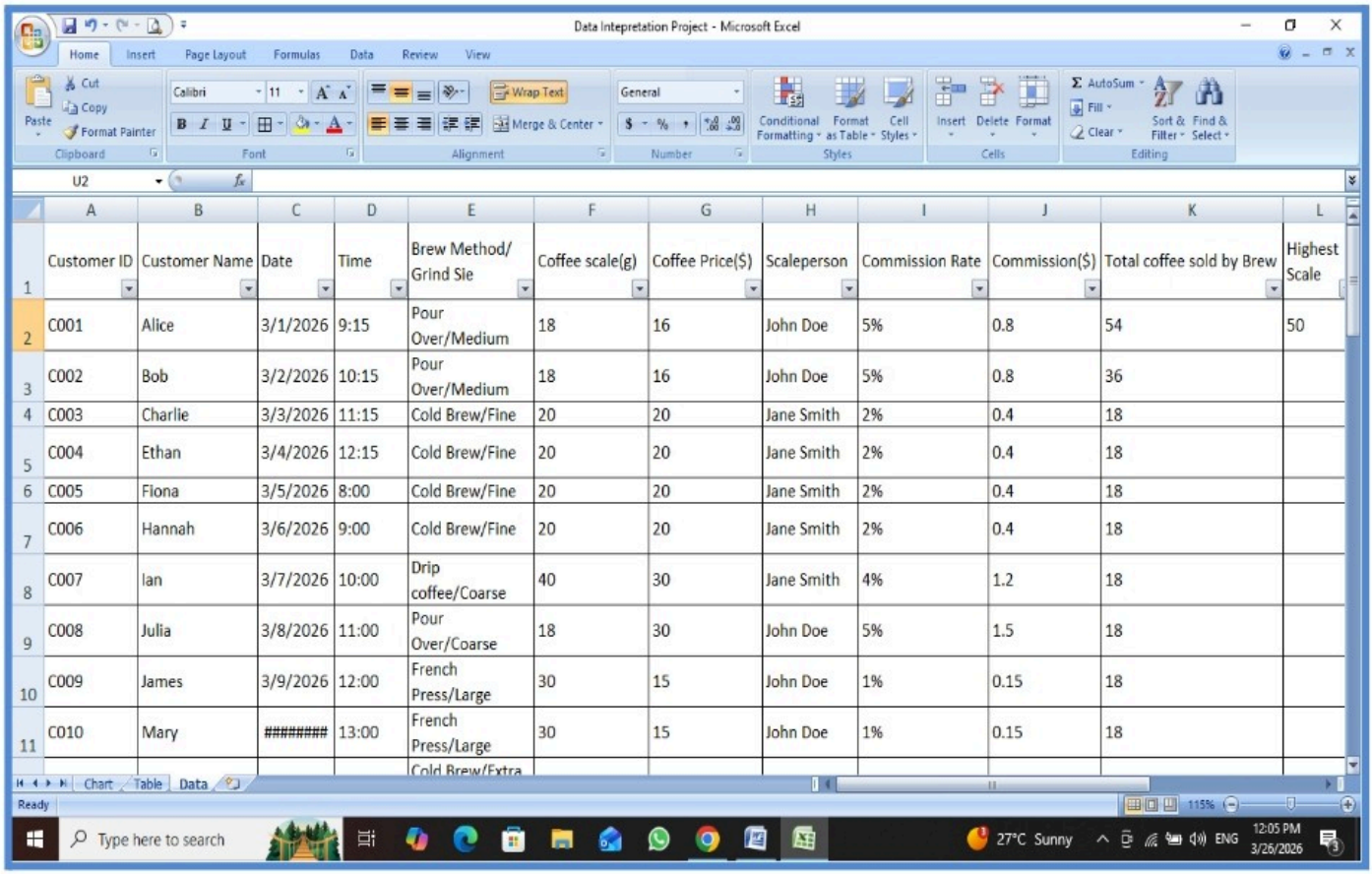
11. Recommendations

* Promote sustainable farming practices
* Use advanced measurement tools for quality control

* Support small farmers with technology and training

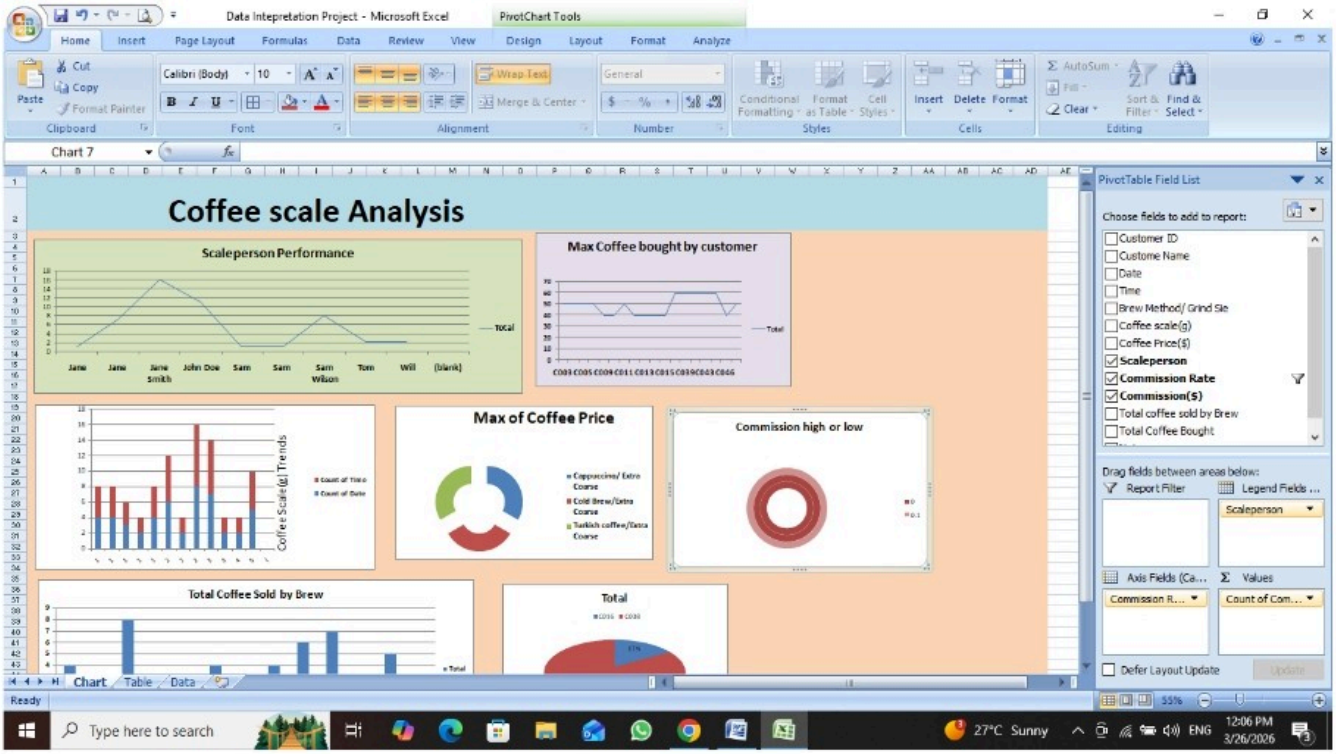
* Encourage fair-trade and ethical sourcing

12.Data in Excel



Customer ID	Customer Name	Date	Time	Brew Method/ Grind Sie	Coffee scale(g)	Coffee Price($)	Scaleperson	Commission Rate	Commission($)	Total coffee sold by Brew	Highest Scale
C001	Alice	3/1/2026	9:15	Pour Over/Medium	18	16	John Doe	5%	0.8	54	50
C002	Bob	3/2/2026	10:15	Pour Over/Medium	18	16	John Doe	5%	0.8	36	
C003	Charlie	3/3/2026	11:15	Cold Brew/Fine	20	20	Jane Smith	2%	0.4	18	
C004	Ethan	3/4/2026	12:15	Cold Brew/Fine	20	20	Jane Smith	2%	0.4	18	
C005	Fiona	3/5/2026	8:00	Cold Brew/Fine	20	20	Jane Smith	2%	0.4	18	
C006	Hannah	3/6/2026	9:00	Cold Brew/Fine	20	20	Jane Smith	2%	0.4	18	
C007	Ian	3/7/2026	10:00	Drip coffee/Coarse	40	30	Jane Smith	4%	1.2	18	
C008	Julia	3/8/2026	11:00	Pour Over/Coarse	18	30	John Doe	5%	1.5	18	
C009	James	3/9/2026	12:00	French Press/Large	30	15	John Doe	1%	0.15	18	
C010	Mary	########	13:00	French Press/Large	30	15	John Doe	1%	0.15	18	

13.Chart/ Output



14.Formula

14.1 Total Coffee Sold by Brew

=SUMIF(E2:E50,"Pour Over/Medium",F2:F50)

14.2 Highest Scale

=MAX(F2:F51)

14.3 Lowest Scale

=MIN(F2:F51)

14.4 Average Scale

=AVERAGE(F2:F51)

14.5 Bonus

=IF(I2>=10,”HIGH”,”LOW”)

14.6 Customer high Scale

=INDEX(A2:A51,MATCH(MAX(F2:F51),F2:F51,0))

14.7 Customer Low Scale

=INDEX(A2:A51,MATCH(MIN(F2:F51),F2:F51,0))

14.8 Highest Coffee Price

= MAX(G2:G51)

14.9 Lowest Coffee Price

=MIN(G2:G51)